

Explainable Deep Learning for Breast Cancer Classification Using Mammography Images

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Abstract—Automated classification of mammograms is difficult because suspicious findings can occupy only a small portion of an image and can resemble normal anatomical structures. Deep neural networks can learn visual representations from these images, although the basis of an individual prediction is not immediately visible. This work evaluates an explainable binary classification framework using the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM). A compact convolutional neural network is used as the reference model and is compared with an ImageNet-initialized EfficientNetB0. Grad-CAM is applied to selected predictions of the reference network to inspect its activation patterns. Dataset cleaning, deduplication, and patient-wise partitioning are performed before model training. The resulting experimental set contains 2,842 distinct images from 1,457 patients, with 443 images reserved for final testing. The baseline CNN obtains 61.85% accuracy, 0.6261 AUC, 65.52% precision, 29.08% malignant recall, and an F1-score of 0.4028. EfficientNetB0 obtains 62.08% accuracy, 0.6670 AUC, 62.28% precision, 36.22% malignant recall, and an F1-score of 0.4581. The transfer-learning model therefore performs better in AUC, malignant recall, and F1-score, while the compact CNN retains the higher precision. Grad-CAM is used only as a qualitative inspection of prediction-related activations. Overall, the experiments show why predictive metrics and explanation behaviour should be considered together when assessing mammographic classifiers.

Index Terms—Breast cancer, mammography, deep learning, convolutional neural network, EfficientNetB0, explainable artificial intelligence, Grad-CAM, CBIS-DDSM.

I. INTRODUCTION

Breast cancer screening relies heavily on mammography, where masses and calcifications may provide important evidence of disease. The visual appearance of these findings is not uniform, and some abnormalities can be difficult to separate from surrounding tissue. Consequently, image interpretation remains a demanding task even when high-quality mammographic examinations are available.

Deep neural networks have been applied to mammographic detection and classification, with CNN-based systems learning image representations directly from training examples [2]–[4]. This feature-learning capability reduces the need to specify a complete set of visual descriptors manually. Pretrained

networks are also attractive for medical imaging experiments because the available task-specific datasets are often much smaller than the datasets used to build modern vision models.

Research using larger screening cohorts has also examined AI systems alongside radiologist assessment [6]–[8]. Such results are encouraging, but performance can depend on the patient population, acquisition protocol, and intended workflow. Another line of work has investigated annotation-efficient learning to reduce the amount of detailed lesion labelling required during development [9].

A separate concern is how to inspect the reasoning represented by a trained network. The output probability alone does not reveal which image structures affected the decision. In medical image analysis this is particularly relevant because researchers need to determine whether the learned response is associated with meaningful visual evidence. XAI techniques can be used to inspect this behaviour after or during model development.

This work evaluates a benign-versus-malignant classification pipeline on CBIS-DDSM [1]. The experimental comparison uses a compact CNN and an EfficientNetB0 transfer-learning model. For the compact CNN, Grad-CAM is generated for representative true-negative, false-positive, false-negative, and true-positive outcomes so that the network's activation behaviour can be inspected visually.

The study is designed around the following objectives:

- to construct a reproducible preprocessing pipeline for binary classification of the CBIS-DDSM images;
- to establish a compact CNN reference classifier;
- to evaluate the reference CNN against an EfficientNetB0 transfer-learning model;
- to inspect selected CNN decisions with Grad-CAM; and
- to interpret the quantitative results together with the limitations of post-hoc visual explanations.

The paper proceeds from prior work in Section II to the dataset and experimental procedure in Section III. Section IV reports the measured results and their interpretation, followed by the conclusion, limitations, and future directions in Section V.

II. RELATED WORK

Mammography has become an active application area for computer-aided deep learning. CBIS-DDSM is frequently used because its image records are accompanied by pathology-related information [1]. Earlier investigations considered lesion detection and classification, followed by studies that moved toward broader screening pipelines [2]–[4]. Other work has compared multiple network designs for mammographic diagnosis [5].

Architectural design has also evolved toward deeper networks and more efficient scaling. Residual connections were introduced to ease the training of deeper CNNs [10]. EfficientNet instead couples changes in depth, width, and input resolution through a compound scaling rule [11]. These architectures motivate the use of pretrained backbones as comparison models in image-classification experiments.

Several large-scale studies have investigated AI-assisted screening and its interaction with radiologist performance [6]–[8]. Their findings also indicate that benchmark scores should be interpreted in the context of population and workflow differences. Work on annotation-efficient training provides another strategy for limiting the reliance on detailed lesion annotations [9].

Interpretation of model decisions is increasingly considered alongside predictive performance in healthcare. Grad-CAM obtains class-dependent activation maps from gradients flowing through convolutional feature maps [12]. LIME and SHAP offer different explanation mechanisms for inspecting model outputs [13], [14]. Recent reviews of breast-imaging XAI point to unresolved issues involving validation, reproducibility, standard evaluation procedures, and clinical use [15], [16]. Mammography-focused evidence also suggests that external testing and clinician-oriented assessment of explanations are still limited [17]. Yadav et al. [24] developed an IoT-based UV sanitization system for package surfaces, demonstrating the use of connected intelligent technologies for automated health and safety applications. In contrast, the present work applies deep learning and explainable AI to mammographic image classification, with Grad-CAM used to provide visual insight into model predictions.

A heatmap that appears visually reasonable should not automatically be interpreted as proof that the network has identified the disease-related structure. Accordingly, Grad-CAM is used here only for qualitative inspection. Its output is discussed together with the numerical test metrics and the EfficientNetB0 comparison, rather than being presented as validated lesion localization.

III. MATERIALS AND METHODS

A. Dataset Description

The experimental data come from the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM). The records include mammographic examinations together with pathology information and regions of interest, making the collection suitable for computer-aided detection and diagnosis studies [1].

TABLE I
IMAGE COUNTS AND CLASS COMPOSITION AFTER DATASET PREPARATION

Subset	Images	Benign	Malignant
Training	1,994	1,113	881
Validation	405	220	185
Testing	443	247	196
Total	2,842	1,580	1,262

Both mass and calcification records were considered during metadata processing, and the associated mammographic file paths were resolved from the supplied identifiers. For the binary task, `exititBENIGN` and `exititBENIGN_WITHOUT_CALLBACK` were mapped to class 0, whereas `exititMALIGNANT` was mapped to class 1.

Repeated image entries were checked before model development. When the same image path occurred with inconsistent labels, that path was excluded instead of forcing one label onto the record. Following this cleaning stage, 2,842 distinct images remained, representing 1,457 patients; 1,580 images were benign and 1,262 were malignant.

B. Patient-Level Data Splitting

The partitioning procedure was performed by patient rather than by individual image. Thus, images belonging to one patient could not be distributed across multiple partitions, reducing the risk that related images would appear in both development and evaluation data.

The resulting partitions comprised 1,994 training images from 1,019 patients, 405 validation images from 219 patients, and 443 testing images from 219 patients. Checks on the patient identifiers showed no overlap among these three groups.

Some patients contributed images carrying both pathology labels. To stratify patients, the maximum image label was used only to assign a patient to a stratum. The individual image labels were left unchanged when constructing the training and evaluation data.

C. Image Preprocessing

Each mammogram was decoded as a grayscale image and resampled to 224×224 pixels. Because both classifiers use three-channel inputs, the single intensity channel was replicated to form an RGB-like representation. Pixel intensities were stored as floating-point values between 0 and 1. Mini-batches contained 32 images; only the training loader used shuffling. The tensor supplied to either network therefore had dimensions $224 \times 224 \times 3$.

D. Classification Framework

The workflow was implemented in four sequential parts: metadata cleaning, image preparation, model inference/training, and explanation analysis. Valid image-label pairs were first assembled from the CBIS-DDSM metadata, after which duplicate and conflicting paths were removed. Patient-wise partitioning was then applied. The two evaluated classifiers were the compact CNN and the frozen-backbone

TABLE II
LAYER CONFIGURATION AND PARAMETER COUNTS OF THE REFERENCE CNN

Layer	Output Shape	Parameters
Conv2D	$224 \times 224 \times 32$	896
MaxPooling2D	$112 \times 112 \times 32$	0
Conv2D	$112 \times 112 \times 64$	18,496
MaxPooling2D	$56 \times 56 \times 64$	0
Conv2D	$56 \times 56 \times 128$	73,856
MaxPooling2D	$28 \times 28 \times 128$	0
Conv2D	$28 \times 28 \times 256$	295,168
MaxPooling2D	$14 \times 14 \times 256$	0
Global Average Pooling	256	0
Dropout	256	0
Dense	128	32,896
Dropout	128	0
Dense (Sigmoid)	1	129
Total	–	421,441

EfficientNetB0. Grad-CAM was generated only for the trained CNN. The experiment was designed to characterize model performance and behaviour, not to claim a clinical diagnostic deployment.

E. Baseline CNN Architecture

The reference network accepts a $224 \times 224 \times 3$ input and stacks four convolutional blocks with 32, 64, 128, and 256 filters. Each block uses a 3×3 convolution with ReLU followed by 2×2 max pooling. Global average pooling compresses the final feature maps. Two dropout stages surround a 128-unit ReLU dense layer, and the final unit is a sigmoid classifier for the malignant class. The complete network has 421,441 trainable parameters.

F. EfficientNetB0 Model

EfficientNetB0 served as the transfer-learning comparator. Its ImageNet-trained feature extractor was not unfrozen during this experiment. The backbone output of $7 \times 7 \times 1280$ was passed through global average pooling, dropout, a 128-unit ReLU layer, another dropout layer, and a sigmoid output. The network used the same $224 \times 224 \times 3$ input size. In total it contains 4,213,668 parameters, with 164,097 trainable parameters.

G. Experimental Setup

For both networks, optimization used Adam at a learning rate of 10^{-4} with binary cross-entropy. Accuracy, AUC, precision, and recall were recorded during training. Training was limited to 10 epochs, and validation AUC determined which checkpoint was retained for the final test evaluation.

For a binary target, the training objective is written as

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (1)$$

Here, y_i denotes the observed class label and $\hat{y}_i$ denotes the network probability assigned to the malignant class.

TABLE III
TEST PERFORMANCE MEASURED FOR THE REFERENCE CNN

Metric	Value
Accuracy	0.6185
AUC	0.6261
Precision	0.6552
Recall	0.2908
F1-score	0.4028

H. Explainable AI Using Grad-CAM

The CNN explanations were generated with Grad-CAM, which uses the gradient of a selected class score with respect to convolutional feature maps [12]. In this work the resulting maps are treated as qualitative evidence of the network's activation pattern.

Let A^k denote feature map k and let y^c be the score for class c . The contribution weight assigned to feature map k is obtained from the spatially averaged gradient:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (2)$$

where Z is the number of spatial positions in the feature map.

The class-specific Grad-CAM activation is subsequently formed as

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (3)$$

After computation, the activation map is normalized, resized to 224×224 , and superimposed on the corresponding mammogram for visual inspection.

The heatmaps are interpreted only as visual indications of areas that contributed to the model response. They are not presented as verified lesion boundaries or clinical localization. Establishing spatial accuracy would require a quantitative comparison against expert or ROI annotations.

IV. RESULTS AND DISCUSSION

A. Baseline CNN Performance

Evaluation on the 443-image test partition gave the baseline CNN an accuracy of 0.6185 and an AUC of 0.6261. Its malignant-class precision was 0.6552, whereas recall was 0.2908; the corresponding F1-score was 0.4028.

An AUC of 0.6261 reflects only modest separation between the two classes. More importantly, the malignant recall of 0.2908 shows that a large fraction of malignant test images were assigned to the benign class.

B. Training Behaviour

Figs. 1–3 summarize the three tracked training curves over the 10 epochs. The training metrics generally moved upward or downward in the expected directions, while the validation curves showed noticeable epoch-to-epoch variation.

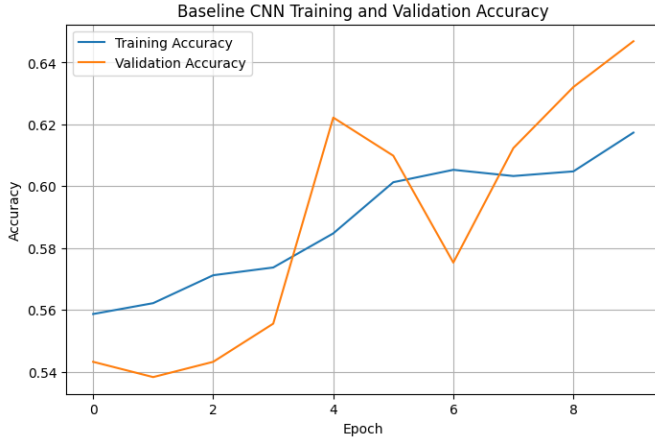


Fig. 1. Accuracy trajectories for the CNN on the training and validation partitions.

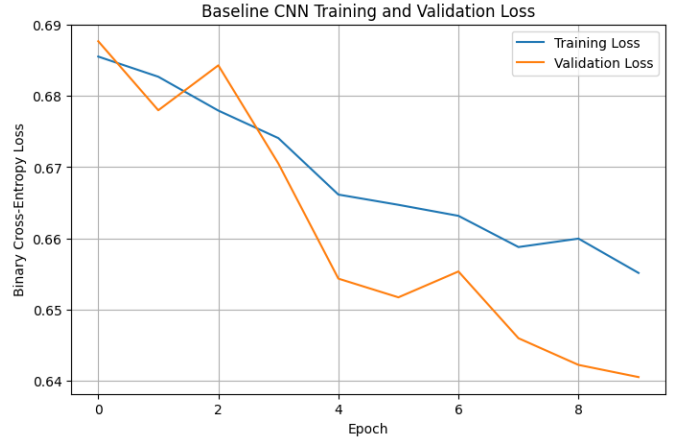


Fig. 3. Binary cross-entropy loss observed for training and validation data over 10 epochs.

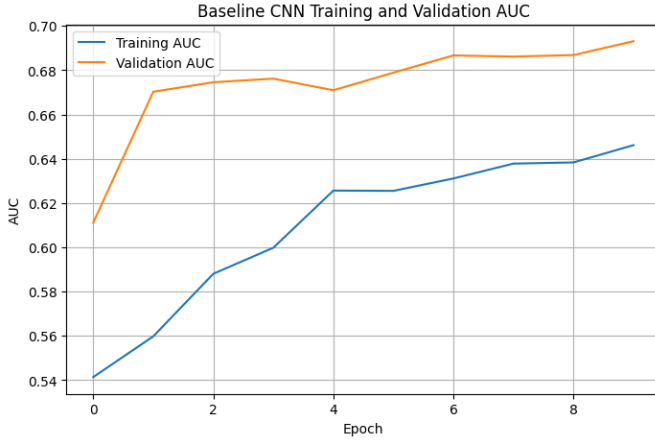


Fig. 2. AUC values recorded on the training and validation partitions during learning.

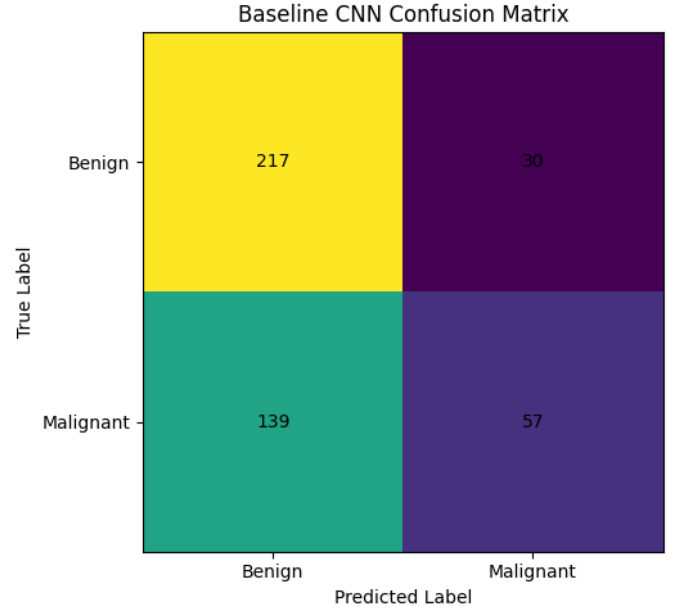


Fig. 4. Test-set prediction counts for benign and malignant classes using the CNN.

C. Confusion Matrix Analysis

Fig. 4 gives the class-wise prediction counts. Of the 247 benign cases, 217 were assigned correctly and 30 were predicted as malignant. For the 196 malignant cases, 57 were detected correctly and 139 were missed by the classifier. These 139 false negatives produce the 29.08% malignant recall and illustrate why the single accuracy value does not capture the full error profile.

D. Grad-CAM Analysis

Four test examples were selected for the Grad-CAM inspection, one from each outcome type: true negative, false positive, false negative, and true positive.

The four heatmap examples are displayed in Fig. 5. The true-negative and true-positive panels show activation patterns for correct decisions, whereas the false-positive and false-negative panels show patterns associated with errors. These examples are intended to illustrate network behaviour rather than to identify pathology directly. A highlighted area may not correspond to the actual abnormality, so spatial validation

against expert annotations would be needed before making stronger claims.

E. Comparison with EfficientNetB0

The performance of the baseline CNN and EfficientNetB0 is summarized in Table IV.

The two models have very similar accuracy values: 0.6208 for EfficientNetB0 versus 0.6185 for the reference CNN. The larger numerical difference appears in AUC, where the transfer-learning model reaches 0.6670 compared with 0.6261.

The transfer-learning model also raised malignant recall from 0.2908 to 0.3622 and F1-score from 0.4028 to 0.4581. In practical terms, a greater share of the malignant test cases received a positive prediction with EfficientNetB0.

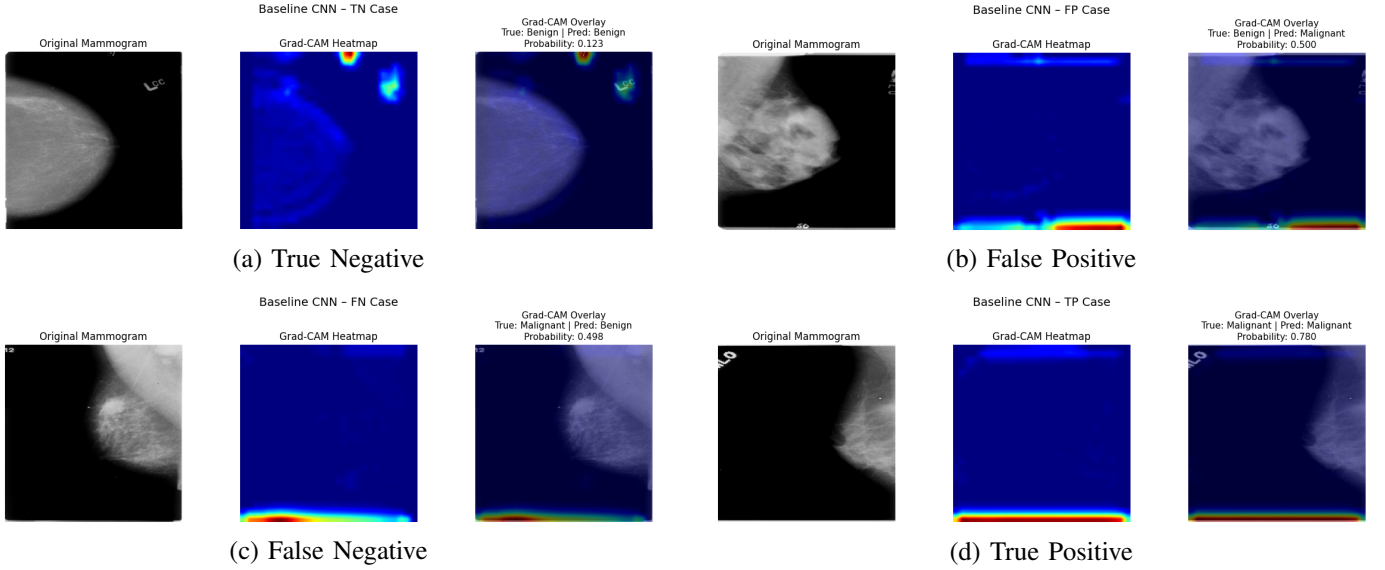


Fig. 5. Grad-CAM examples corresponding to four CNN outcomes: true negative, false positive, false negative, and true positive.

TABLE IV
METRIC COMPARISON BETWEEN THE REFERENCE CNN AND EFFICIENTNETB0

Model	Accuracy	AUC	Precision	Recall	F1-score
Baseline CNN	0.6185	0.6261	0.6552	0.2908	0.4028
EfficientNetB0	0.6208	0.6670	0.6228	0.3622	0.4581

This gain in recall came with a reduction in precision. EfficientNetB0 recorded 0.6228 precision, compared with 0.6552 for the baseline CNN. The comparison is therefore a metric trade-off rather than an across-the-board improvement.

Considering the complete metric set, EfficientNetB0 has the advantage in AUC, malignant recall, and F1-score, whereas the baseline network has the better precision. Since the accuracy difference is small, accuracy alone would give an incomplete view of this comparison.

V. CONCLUSION

The study evaluated an explainable mammographic classification workflow using CBIS-DDSM. Following metadata cleaning and deduplication, 2,842 images from 1,457 patients were retained, and patient-wise partitioning was used for training, validation, and final testing.

On the held-out test data, the reference CNN reached 61.85% accuracy, 0.6261 AUC, 65.52% precision, 29.08% malignant recall, and 0.4028 F1-score. EfficientNetB0 reached 62.08% accuracy, 0.6670 AUC, 62.28% precision, 36.22% malignant recall, and 0.4581 F1-score. Thus, the transfer-learning model offered better class separation and malignant sensitivity, while the smaller CNN produced fewer false positives among its positive predictions.

Grad-CAM was used to inspect activation patterns for the four selected prediction outcomes. The resulting examples give a qualitative picture of how the CNN responds to different test cases. They should not, however, be interpreted as clinically validated lesion maps.

A. Limitations

Several constraints should be considered when interpreting these results. The experiments use only CBIS-DDSM, so performance may change for images acquired at other institutions or with different imaging systems and patient populations.

The malignant recall of both models is also modest, which means that many malignant cases remain undetected. This is an important weakness for any application in which missed positive cases carry substantial consequences.

The explanation study is qualitative rather than quantitatively validated. The Grad-CAM maps were not scored against expert lesion boundaries, so they describe network activations without establishing clinical localization accuracy.

In addition, the EfficientNetB0 feature extractor remained frozen. Fine-tuning under a controlled experimental protocol could change the observed comparison.

B. Future Work

Future experiments should concentrate on raising malignant-case sensitivity without causing an excessive increase in false positives. Useful directions include stronger augmentation, explicit treatment of class imbalance, and systematic hyperparameter searches.

The pretrained EfficientNetB0 backbone can also be fine-tuned under carefully controlled settings. Testing on an independent mammography dataset would provide additional evidence about cross-dataset generalization.

For XAI, the available ROI information could be used to quantify the spatial agreement of Grad-CAM maps with annotated regions. Comparing several explanation methods would further indicate whether the observed patterns are stable across XAI techniques.

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